

Cramer's

$$\begin{aligned} 1x + 2y + 3z &= 18 \\ 2x + 0y + 4z &= 6 \\ 0x - 2y - 7z &= 7 \end{aligned}$$

note!

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 4 \\ 0 & -2 & -7 \end{bmatrix}$$

$$A_x = \begin{bmatrix} 18 & 2 \\ 6 & 0 \\ 7 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 4 \\ 0 & -2 & -7 \end{bmatrix} \quad \begin{matrix} x_1 & x_2 & x_3 \\ \begin{bmatrix} 5 & 5 & 5 \\ 5 & -5 & 0 \\ 5 & 0 & 5 \end{bmatrix} \end{matrix}$$

$$\begin{aligned} 5x + 5y + 5z \\ 5x - 5x \\ 5x + 5z \end{aligned}$$

$$x_1 = \frac{|A_1|}{|A|} \quad \text{cramer's rule!}$$

Gauss's Rule

note!

$$\begin{aligned} x + 2y + 5z &= 18 \\ 2x + 0y + 4z &= 6 \\ 9x + 3y &= 12 \end{aligned} \quad \begin{bmatrix} 1 & 2 & 5 \\ 2 & 0 & 4 \\ 9 & 3 & 0 \end{bmatrix}$$

1) สามารถสลับแถวไหนก็ได้

$$\begin{aligned} \text{ex. } R_1 &\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} R_2 \\ R_2 &\begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} R_1 \\ R_1 &\leftrightarrow R_2 \end{aligned} \quad \begin{bmatrix} R_1 & \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix} \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 3 & 3 \\ 2 & 2 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

2) ค่าคงที่สามารถคูณเข้าไปในแถวใดก็ได้ ($\neq 0$)

$$\begin{aligned} \text{ex. } R_1 &\begin{bmatrix} 2 & 16 \\ 2 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 8 \\ 2 & 8 \end{bmatrix} \\ R_2 &\begin{bmatrix} 2 & 8 \\ 2 & 8 \end{bmatrix} \\ (\frac{1}{2}) R_1 &\rightarrow R_1 \end{aligned}$$

3) ค่าคงที่คูณสักแถวหนึ่ง แล้วเอามาทำอีกแถว

$$\begin{aligned} \text{ex. } R_2 &\begin{bmatrix} 2 & 16 \\ 2 & 8 \end{bmatrix} \\ R_2 &- (\frac{1}{2}) R_1 \rightarrow R_2 \end{aligned}$$

$$\begin{bmatrix} R_1 & \begin{bmatrix} 1 & 8 \\ 2 & 8 \end{bmatrix} \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 8 \\ 2 & 8 \end{bmatrix}$$

$$\text{ลบกัน} \quad \begin{bmatrix} \begin{bmatrix} 3 & 6 & 9 \\ 7 & 8 & 12 \\ 6 & 12 & 18 \end{bmatrix} \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 6 & 9 \\ 7 & 8 & 12 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_3 - 2R_1 \rightarrow R_3$$

Gauss's Jordan

$$x - 2y + 3z = 9$$

$$-x + 3y = -4$$

$$2x - 5y + 5z = 17$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ -1 & 3 & 0 & -4 \\ 2 & -5 & 5 & 17 \end{array} \right]$$

$$R_2 + (-1)R_1 \rightarrow R_2$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 2 & -5 & 0 & 17 \end{array} \right]$$

$$R_3 + (-2)R_1 \rightarrow R_3$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & -1 & -1 & -1 \end{array} \right]$$

$$R_3 + R_2 \rightarrow R_3$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & 4 \end{array} \right]$$

$$\left(\frac{1}{2}\right)R_3 \rightarrow R_3$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 9 \\ -1 & 3 & 0 & -4 \\ 2 & -5 & 5 & 17 \end{array} \right]$$

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$$\left[\begin{array}{ccc|c} 1 & 12 & 6 & 8 \\ 0 & 1 & 21 & 16 \\ 0 & 0 & 1 & 7 \end{array} \right]$$

$$x + 12y + 6z = 8$$

$$+ y + 21z = 16$$

$$z = 7$$

$$\begin{bmatrix} 1 & -2 & 3 & 9 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix} \div 7 \quad \begin{array}{l} x - 2y + 3z = 9 \quad - \textcircled{1} \\ y + 3z = 5 \quad - \textcircled{2} \\ z = 2 \end{array}$$

$$\begin{array}{l|l} \text{b69796 } z=2 \text{ g4 } \textcircled{2} & \text{b69796 } z=2, y=-1 \text{ g4 } \textcircled{1} \\ y + 3(2) = 5 & x + 2 + 6 = 9 \\ y = -1 & x = 1 \end{array}$$

$$\therefore z=2, y=-1, x=1$$

Jordan

$$1x_1 + 3x_4 = 4$$

$$6x_2 - 3x_3 - 3x_4 = 0$$

$$3x_2 - 2x_4 = 1$$

$$2x_1 - x_2 + 4x_3 = 5$$

$$\begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 6 & -3 & -3 \\ 0 & 3 & 0 & -2 \\ 2 & -1 & 4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$\rightarrow [A|B] \begin{bmatrix} 1 & 0 & 0 & 3 & 4 \\ 0 & 6 & -3 & -3 & 0 \\ 0 & 3 & 0 & -2 & 1 \\ 2 & -1 & 4 & 0 & 5 \end{bmatrix}$$

$$2006$$

$$2R_1 - R_4 \rightarrow R_4$$

$$[A|B] \left[\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 4 \\ 0 & 6 & -3 & -3 & 0 \\ 0 & 3 & 0 & -2 & 1 \\ 0 & 1 & -4 & 6 & 3 \end{array} \right]$$

$$\frac{R_2}{3} \rightarrow R_2 \left[\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 4 \\ 0 & 2 & -1 & -1 & 0 \\ 0 & 3 & 0 & -2 & 1 \\ 0 & 1 & -4 & 6 & 3 \end{array} \right]$$

$$R_2 \leftrightarrow R_4 \left[\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 4 \\ 0 & 1 & -4 & 6 & 0 \\ 0 & 3 & 0 & -2 & 1 \\ 0 & 2 & -1 & -1 & 3 \end{array} \right]$$

$$3R_2 - R_3 \rightarrow R_3$$

$$03 - 12 \ 18 \left[\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 4 \\ 0 & 1 & -4 & 6 & 0 \\ 0 & 0 & -12 & 20 & 8 \\ 0 & 2 & -1 & -1 & 3 \end{array} \right]$$

$$2R_2 - R_4 \rightarrow R_4$$

$$02 - 8 \ 12 \left[\begin{array}{cccc|c} 1 & 0 & 0 & 3 & 4 \\ 0 & 1 & -4 & 6 & 0 \\ 0 & 0 & -12 & 20 & 8 \\ 0 & 0 & -7 & 13 & 6 \end{array} \right]$$